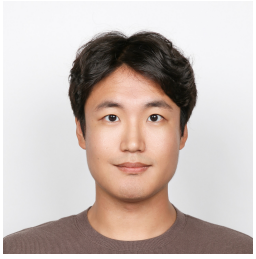
Joonhee Kim, Ph.D. Candidate

Physical Battery Digital Twins & AI-Driven Battery Management

junheemike@postech.ac.kr ☎ +82 10-8312-8458



Education

2021. 03. – 2026. 02. **Ph.D., POSTECH,** Convergence IT Engineering.
Thesis title: *Physical Digital Twin Framework for Advanced Battery Management.*

2016. 03. – 2021. 02. **B.S., POSTECH,** Mechanical Engineering.

2018. 08. – 2019. 01. **Exchange Student, Grenoble INP,** Energy and Smart Systems.

Employment History

2018. 03. – 2018. 08 **Intern.** Eightpercent Co., Ltd.

Research Publications

Journal Articles

1. **J. Kim** et al., "A physics-driven generative model to accelerate artificial intelligence development for lithium-ion battery diagnostics," *Applied Energy*, vol. 391, p. 125 873, 2025.
2. D. Lim, C. Park, **J. Kim**, J. Hong, and S. Han, "PIT-NBV: Poisson-informed transformer for 6-DOF next best view planning in 3D object reconstruction with narrow field of view," *IEEE Transactions on Cognitive and Developmental Systems*, 2025.
3. H. Pyeon et al., "Ultrafast computation of lithium-ion battery dynamics using its closed-form solutions," *Journal of Energy Storage*, vol. 135, p. 117 773, 2025.
4. **J. Kim**, H. Chun, H. Kim, M. Lee, and S. Han, "Strategically switching metaheuristics for effective parameter estimation of electrochemical lithium-ion battery models," *Journal of Energy Storage*, vol. 64, p. 107 094, 2023.

Conference Proceedings

1. J. Choi, **J. Kim**, J. Lee, S. Kim, S. Lee, and S. Han, "Information-theoretic approach to optimal experimental design of current profiles for accurate battery parameter estimation," in *Electrochemical Society Meeting Abstracts 248*, The Electrochemical Society, Inc., 2025, pp. 3446–3446.
2. D. KIM, **J. Kim**, K. Yoon, S. Kim, S. Lee, and S. Han, "Interpretable and lightweight SOH estimation for Li-ion battery from EIS via bandwise CNN-attention," in *Electrochemical Society Meeting Abstracts 248*, The Electrochemical Society, Inc., 2025, pp. 3436–3436.
3. H. Moon, **J. Kim**, and S. Han, "Maximizing the performance of data-driven capacity estimation for lithium-ion battery," in *IFAC-PapersOnLine*, vol. 58, Elsevier, 2024, pp. 31–37.
4. K. Yoon, **J. Kim**, H. Pyeon, K. Kim, and S. Han, "Physics-informed neural network for heat transfer problem with cylindrical heat sources in battery pack geometry," in *2024 24th International Conference on Control, Automation and Systems (ICCAS)*, IEEE, 2024, pp. 702–705.

Patents

KR

- Myeongjae Lee, **Joonhee Kim,** Soohee Han, Jungsu Kim, Jinho Kim, Hangyeol Kim, Hyosik Moon, Hwiyong Jeon, "METHOD AND APPARATUS FOR TRAINING SHORT CIRCUIT DETECTION MODEL," Korean Registered Patent No. 10-2877564-0000, registered Oct. 23, 2025.
- Jungsu Kim, **Joonhee Kim,** Jinho Kim, Myeongjae Lee, Hwiyong Jeon, Soohee Han, "METHOD AND DEVICE WITH BATTERY MODEL OPTIMIZATION," Korean Patent Application Publication No. 10-2024-0036437, published Mar. 20, 2024.

US

- Myeongjae Lee, **Joonhee Kim,** Soohee Han, Jungsu Kim, Jinho Kim, Hangyeol Kim, Hyosik Moon, Hwiyong Jeon, "METHOD AND APPARATUS FOR TRAINING SHORT CIRCUIT DETECTION MODEL," US Patent Application Publication US 2025/0124346 A1, published Apr. 17, 2025.
- Jungsu Kim, **Joonhee Kim,** Jinho Kim, Myeongjae Lee, Hwiyong Jeon, Soohee Han, "METHOD AND DEVICE WITH BATTERY MODEL OPTIMIZATION," US Patent Application Publication US 2024/0111921 A1, published Apr. 4, 2024.

Projects

2025 – · · · ·

- **Hyundai Mobis,** Development of Big-Data-Driven Predictive Models for Degradation Factors in Battery Systems.
- **Samsung Electronics,** Optimization of Fast-Charging Current Protocols using Advanced Battery Degradation Models.

2023 – 2024

- **Samsung Electronics,** Optimization of Fast-Charging Current Profiles for Reducing Lithium-Ion Battery Degradation Using Reinforcement Learning.

2022 – 2023

- **Samsung Electronics,** Development of AI-Based Battery Short-Circuit Detection Algorithms Using Virtual Battery Models.

2021 – 2024

- **National Research Foundation of Korea,** Development of Electrochemical Analysis Tools for Lithium-Ion Batteries Considering Morphological Features.

2021 – 2022

- **Samsung Advanced Institute of Technology,** Development of Physics-Based Electrochemical Models for Internal Short Circuits in Batteries.

Awards & Honors

2025

- **Honorable Mention**, "Introducing My Research" 3-Minute Science Talk Competition, POSTECH

2021

- **Excellence Award** (CEO Award, POSTECH Holdings), POSTECH Startup Competition